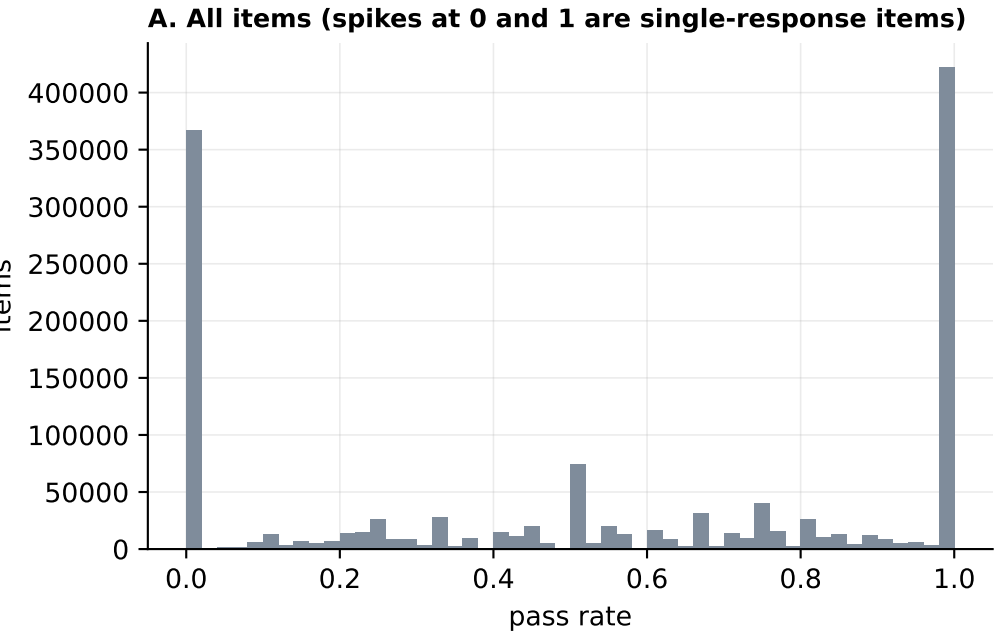
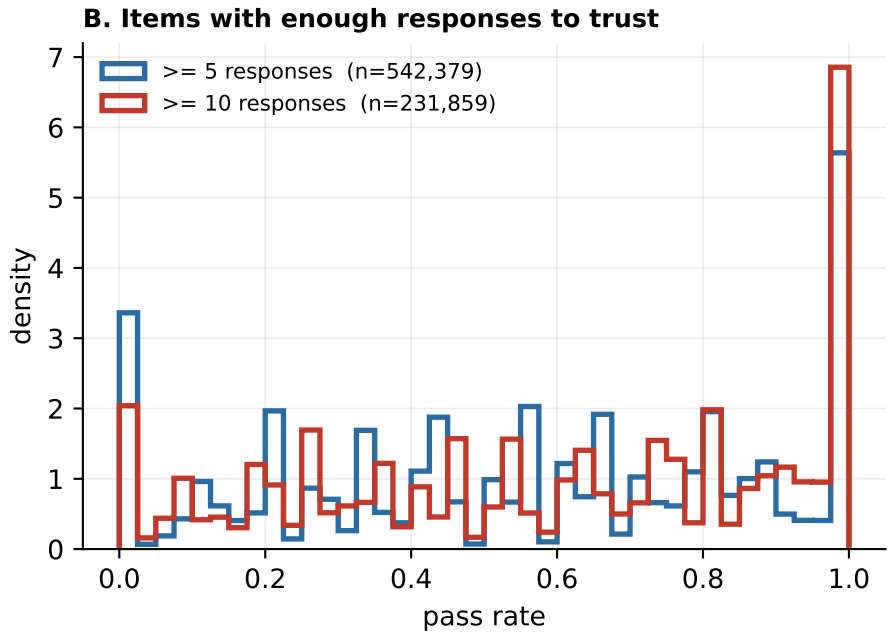
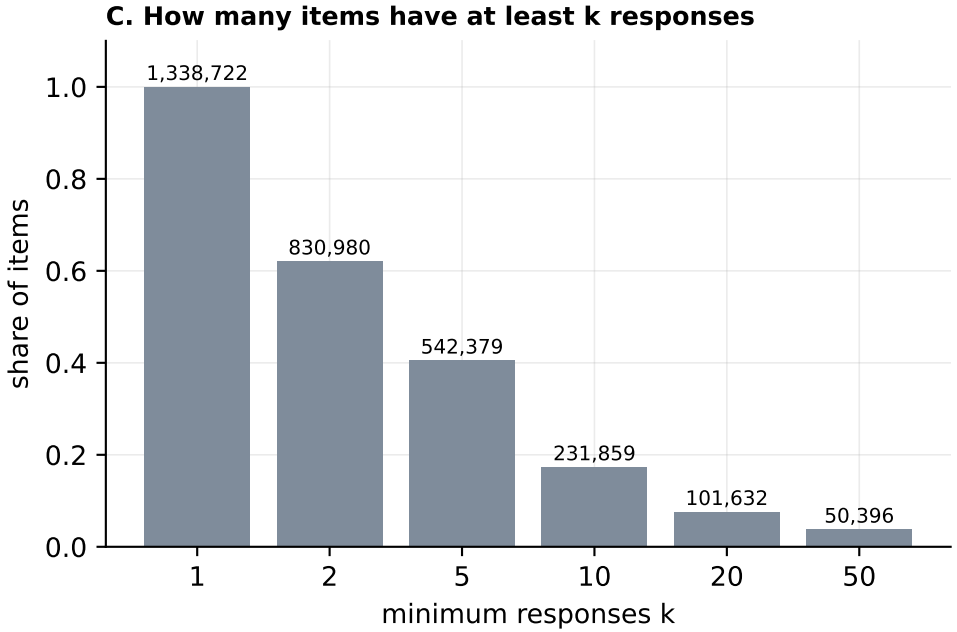
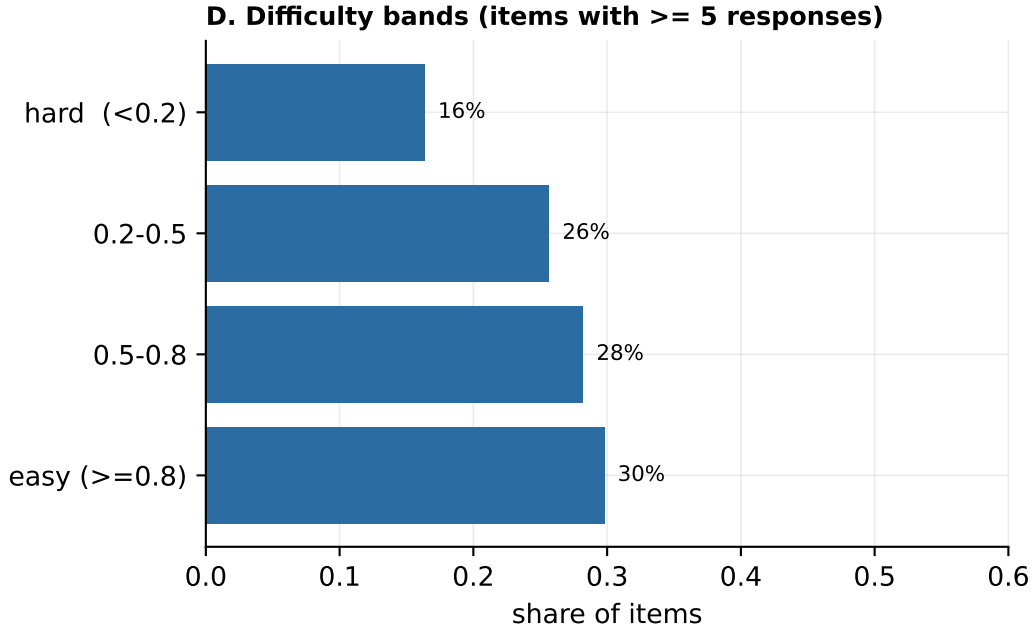
Item pass rate, all benchmarks together

1,338,722 items with at least one response. Pass rate = share of responses that were correct (low = hard). No splits used.





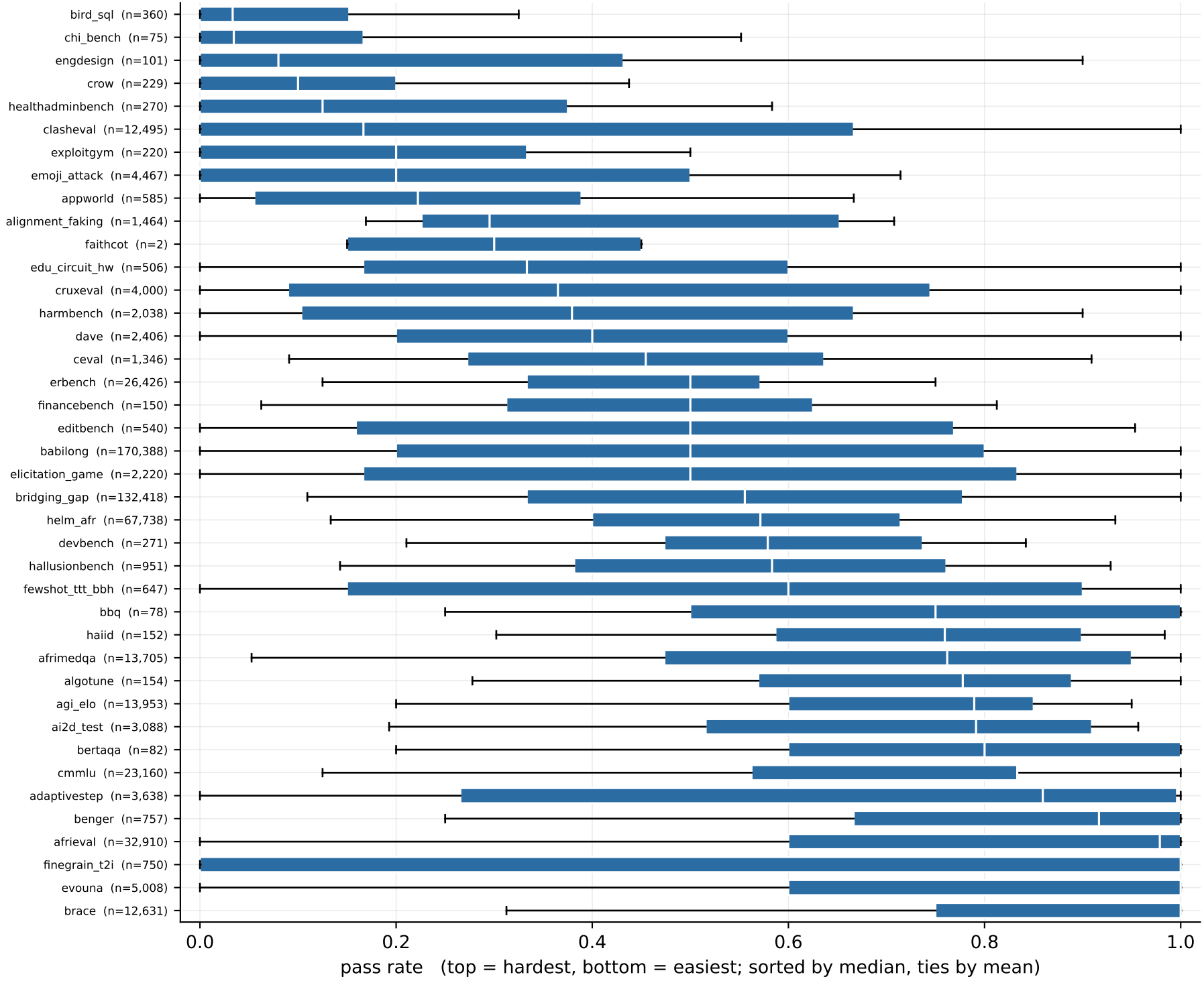




Median pass rate 0.57; middle 50% of items between 0.29 and 0.82; 5th-95th percentile 0.00-1.00.

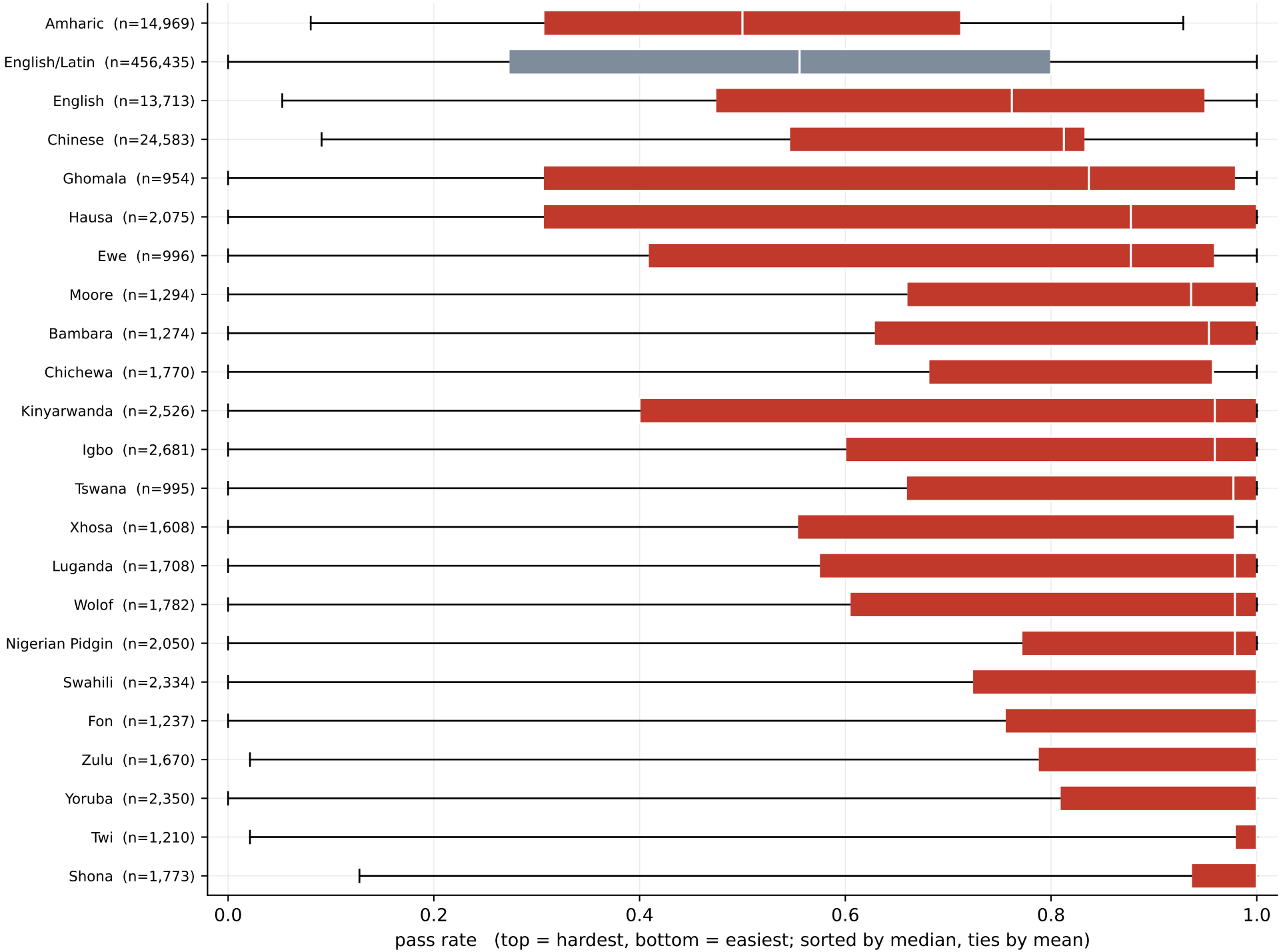
Item pass rate per benchmark

Items with >= 5 responses. Box = middle 50%, line = median, whiskers 5th-95th percentile. Label shows qualifying items.



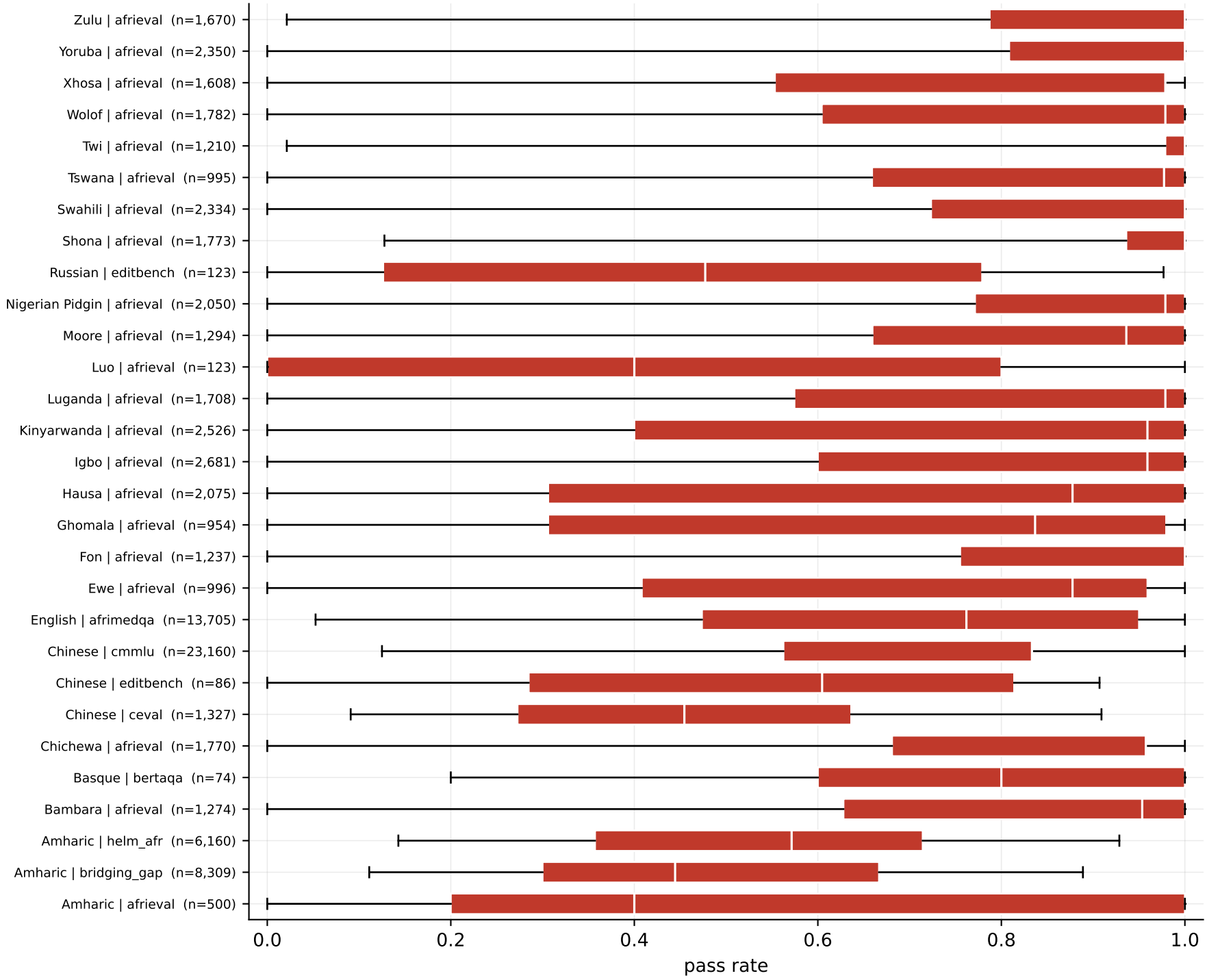
Item pass rate per language (all benchmarks pooled)

Items with >= 5 responses; languages with >= 150 such items. Language is a heuristic: a lang= tag, else dominant non-Latin script, else English/Latin.



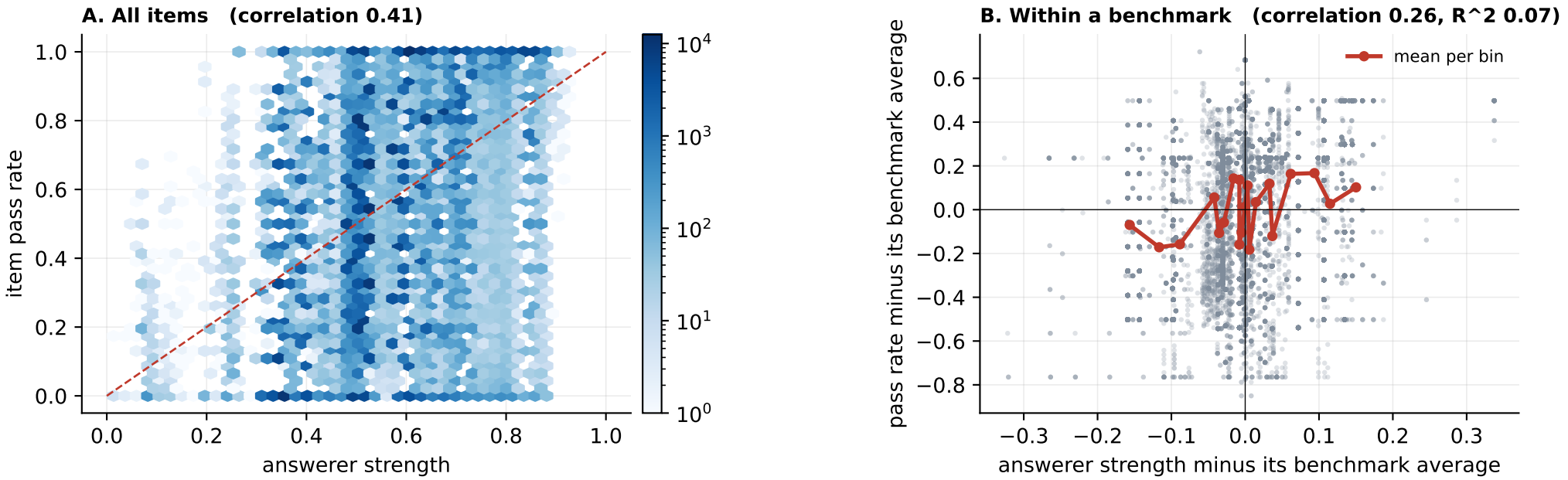
Item pass rate per language per benchmark

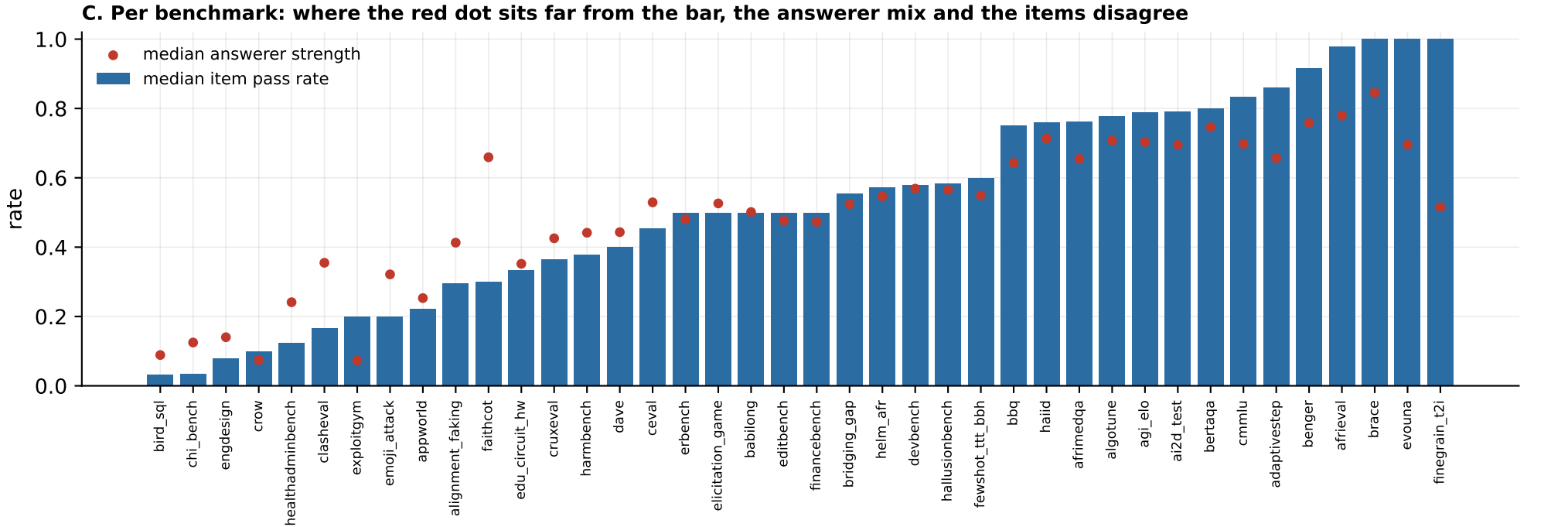
Non-English cells with >= 50 items having >= 5 responses. Most languages live in a single benchmark, so language and benchmark are tangled.



How much of an item's pass rate is just who answered it?

Answerer strength = average accuracy of the subjects that answered the item, measured on their OTHER items (the item cannot predict itself).
Items with >= 5 responses, n=542,379. Colour in A = number of items, log scale.





The 128-item ceiling is not a small trim

It is 128 items inside every subject × benchmark training task — and it happens upstream.

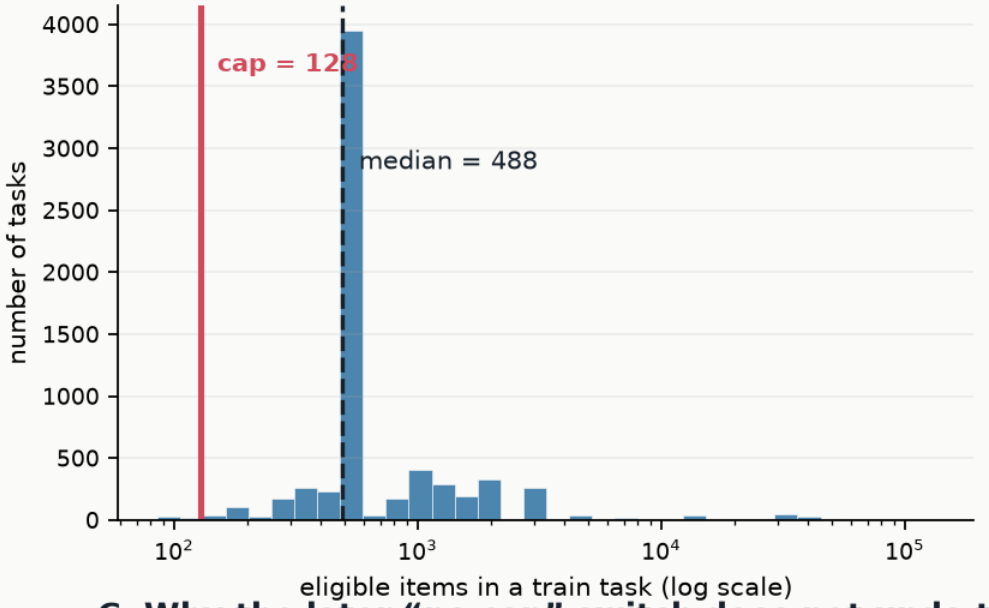
10.82M
eligible task–item rows before the ceiling

856,947
rows available to training after the ceiling

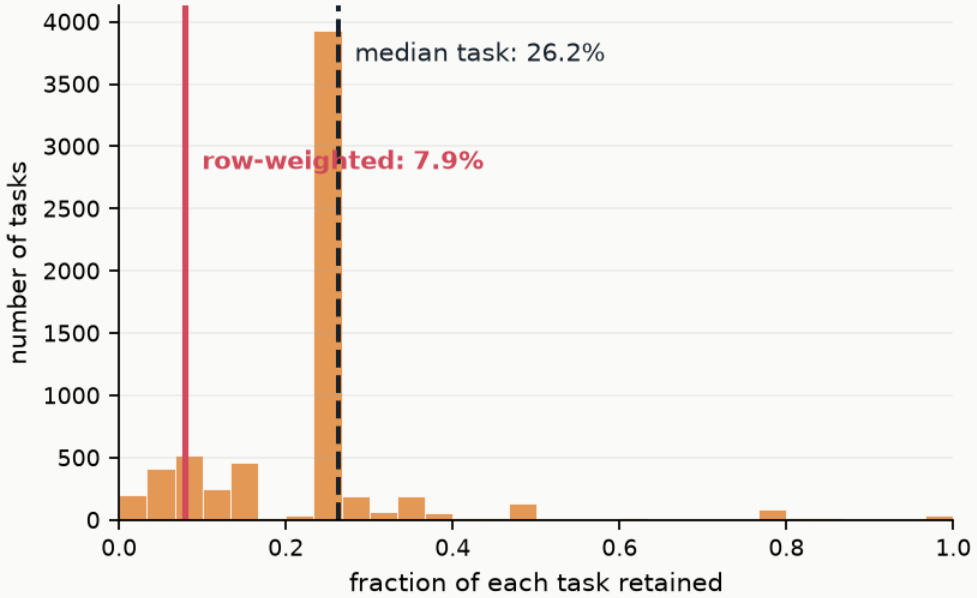
7.9%
of eligible rows retained overall

99.4%
of train tasks end at exactly 128 items

A. Most tasks contain far more than 128 items



B. Retention varies sharply by task size



C. Why the later “no cap” switch does not undo this

Eligible protocol
10,823,690 train rows
median task = 488 items

→

allbench-features-v1
856,947 rows
head(128) per task

→

compact mE5 store
856,947 rows
cap=0 still reads this source

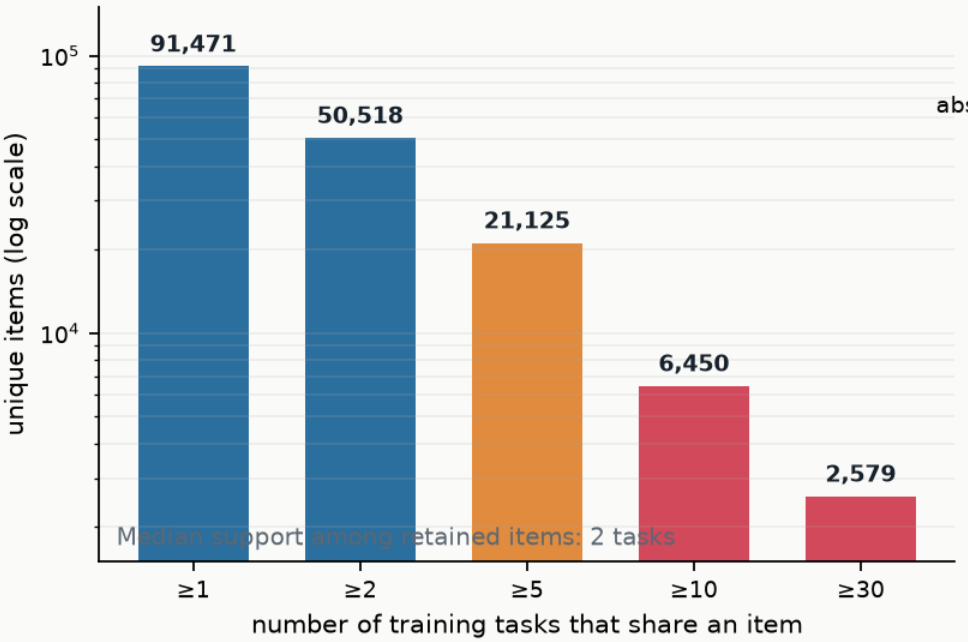
The second store can only remove more rows; it cannot restore item rows already absent from its input.

Exact audit of the current artifacts: 6,702 train tasks; min/median/max kept = 85 / 128 / 128.

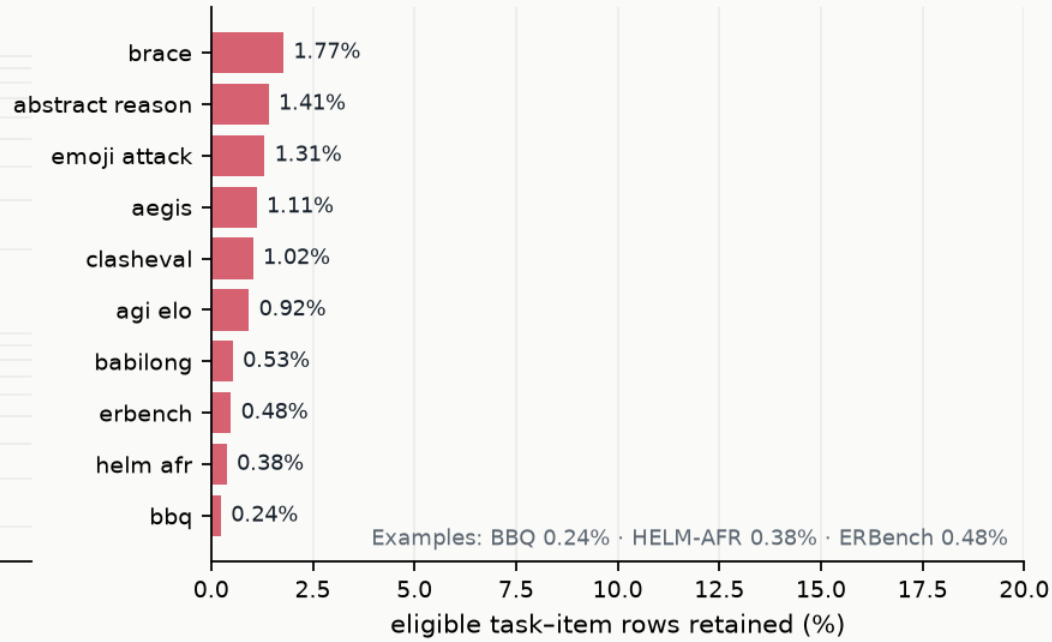
The cap weakens item-difficulty supervision

Difficulty needs the same item answered by multiple subjects. Independent 128-item samples reduce that overlap.

A. Very few retained items have broad supervision



B. Large benchmarks are almost erased



C. Expected overlap between two subjects from the same benchmark

If each task independently keeps 128 of N items, expected shared items ≈ 128^2 / N.

Typical task

N = 488

33.6

expected shared items

BRACE median

N = 7,661

2.1

expected shared items

BBQ median

N = 58,414

0.28

expected shared items

This is the mechanism: less cross-subject overlap → less direct evidence for transferable item difficulty.

Support panel uses the actual capped learner corpus. Benchmark retention is exact for eligible train tasks.

Preprocessing risks beyond the 128 cap

Several data choices can independently flatten or mismeasure item transfer. These should be audited before another model change.

Priority	Current behavior	Why it matters
HIGH	**Repeated task × item rows** 16.28M response rows vs 11.80M distinct prediction inputs. Training keeps the first observation; raw pass-rate plots count all rows.	The learner label and the plotted difficulty target are not defined from the same rows.
HIGH	**Exact duplicate item content** 43.2% of item rows repeat exact content under separate item IDs. There is no content-level deduplication before the current corpus.	Duplicates alter benchmark weighting and can create cross-split near-copies.
HIGH	**Sparse item support** No minimum subject-count filter. About 38% of train items and 55% of test items are single-subject estimates.	A 0/1 pass rate from one subject is mostly noise, not stable difficulty.
HIGH	**Test diagnostic coverage** Only 2,488 of 90,463 test items have ≥5 subjects; all of them come from just DAVE and BERTAQA.	The plotted “test transfer ≈ 0” does not represent all six test benchmarks.
CHECK	**Response-pattern filtering** No filter removes all-correct, all-wrong, or identical-response-vector items.	Numerical stability and effective sample weighting may differ by benchmark.
OK	**Missing answers** Missing outcomes are dropped from observations and therefore masked, not filled.	This part is behaving as intended.
DIFFERENT	**Generalization split** Whole benchmarks are held out. There is no random 20% cell mask and no separate whole-subject holdout in this protocol.	Valid for the competition question, but not comparable to every Stanford diagnostic.

Bottom line: remove the cap, but audit rows, duplicates, support, and coverage together.